Imitation-Guided World Models for Multi-Agent Train Rescheduling

Max Bourgeat, Antoine Legrain, Quentin Cappart

CPAIOR 2026

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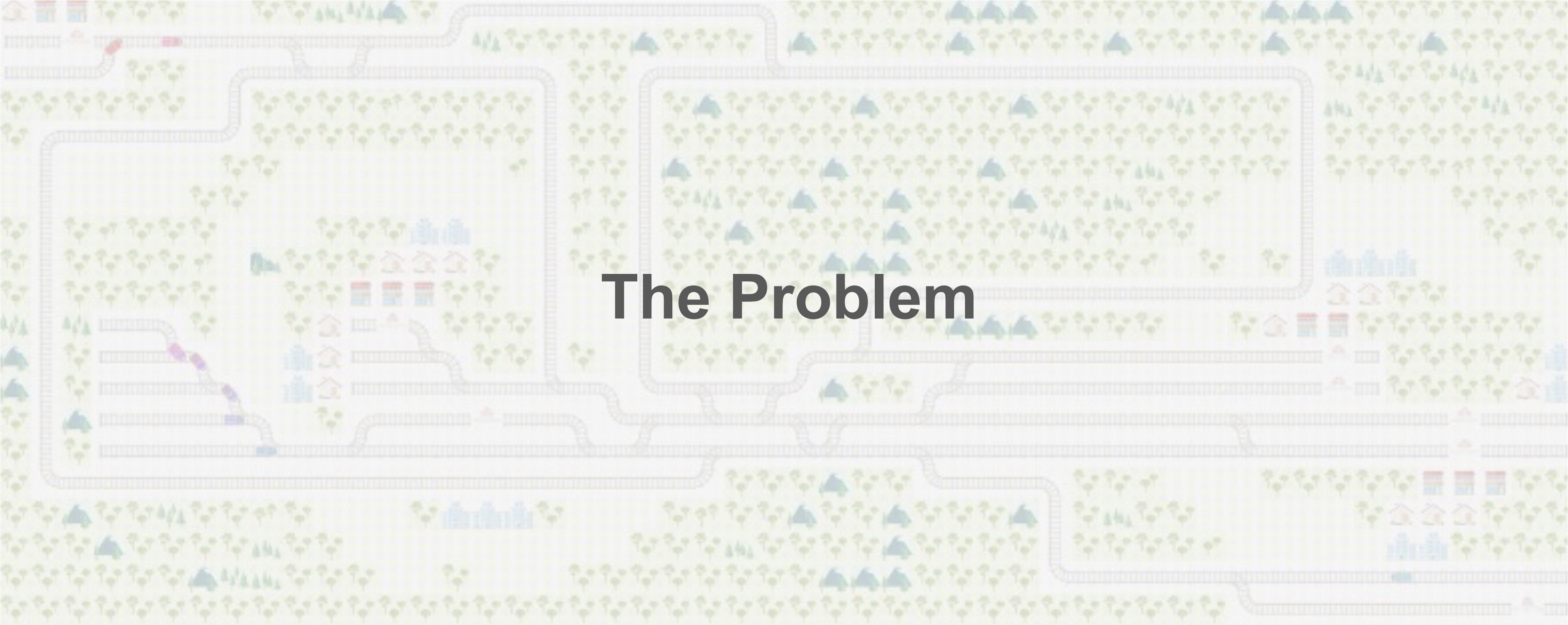


The Problem

Railway environment

Goal : "Operational railway decision support based on reinforcement learning"

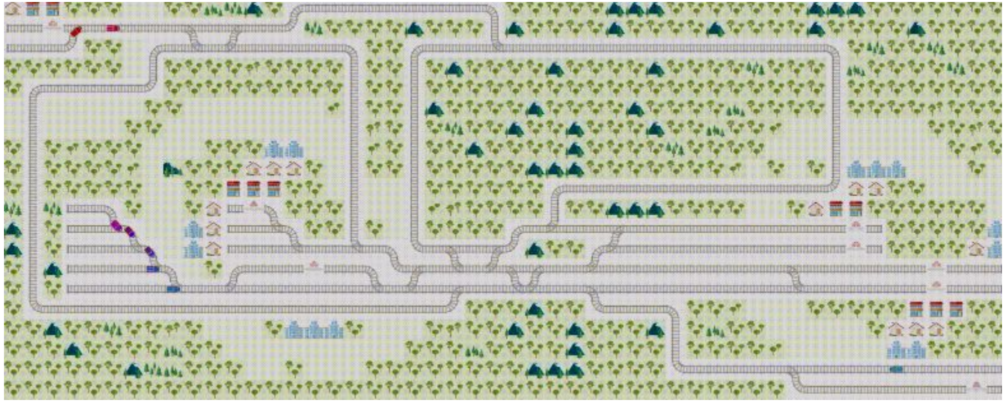
Railway environment : dynamic/stochastic, complex, "high-consequence environment"

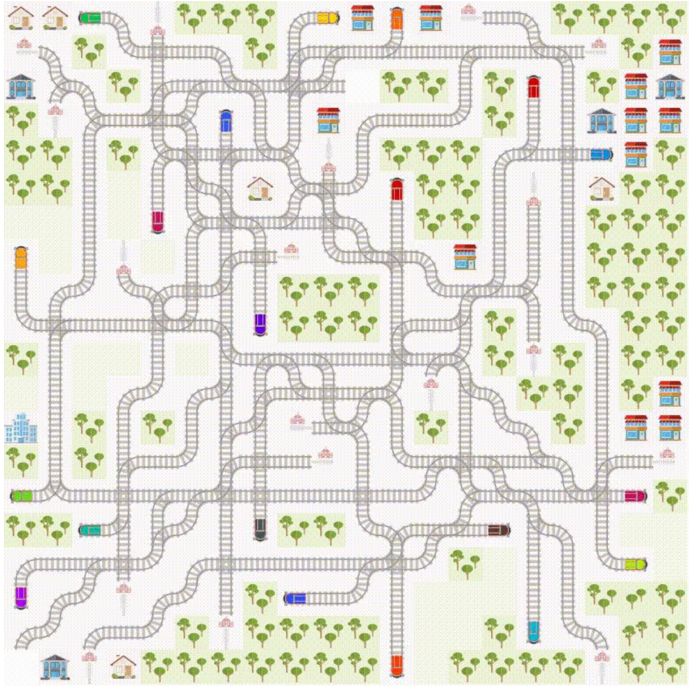
Upstream decision-making : train routing planning

Real-time decision-making : train routing re-optimization

Reinforcement learning : learning paradigm based on trial and error

Railway environment





SBB CFF FFS

SNCF

DB

Flatland : Simulation environment of a railway network

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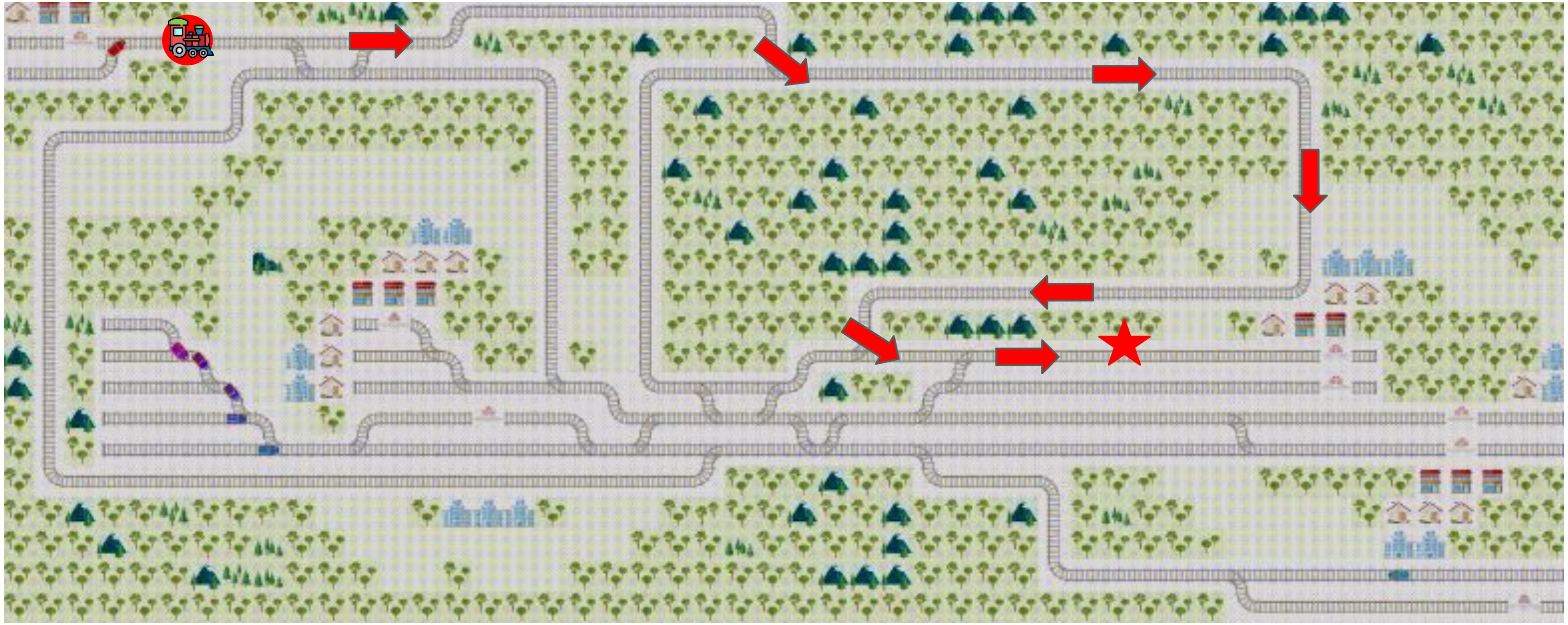
Worst nightmare = Deadlocks



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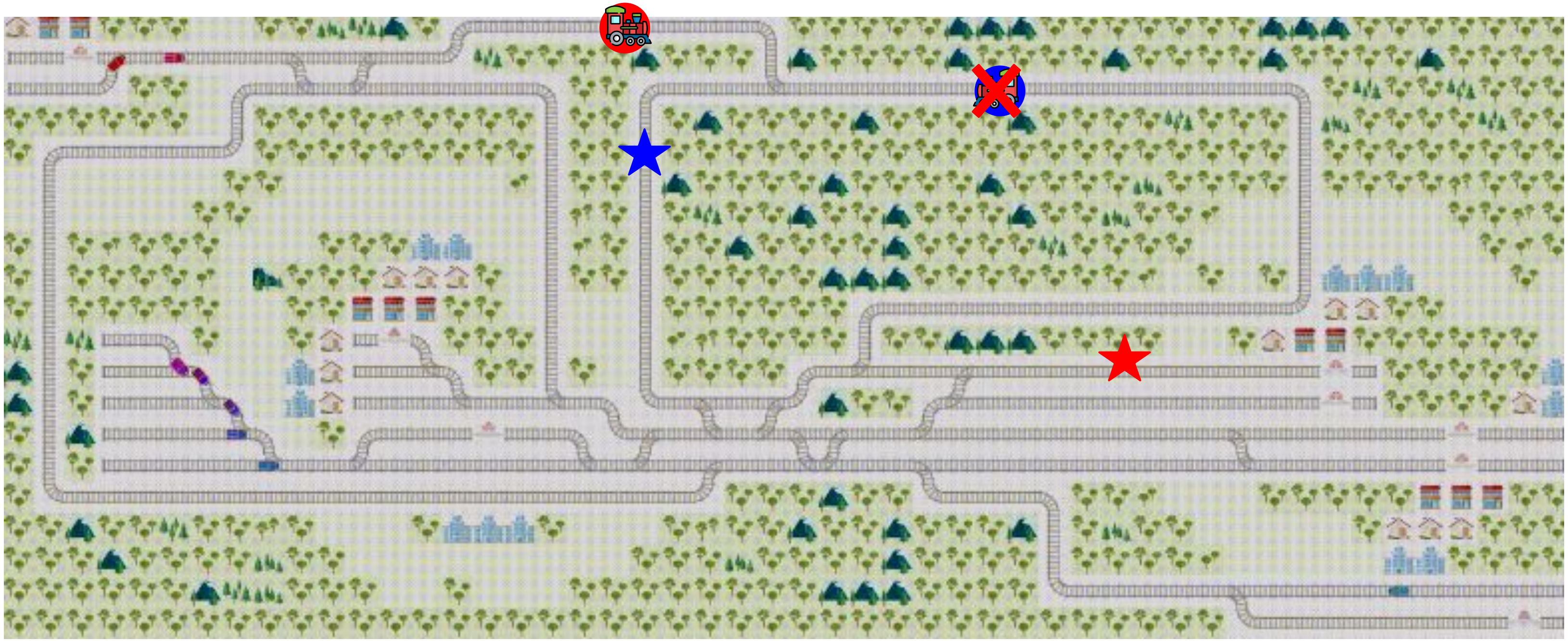

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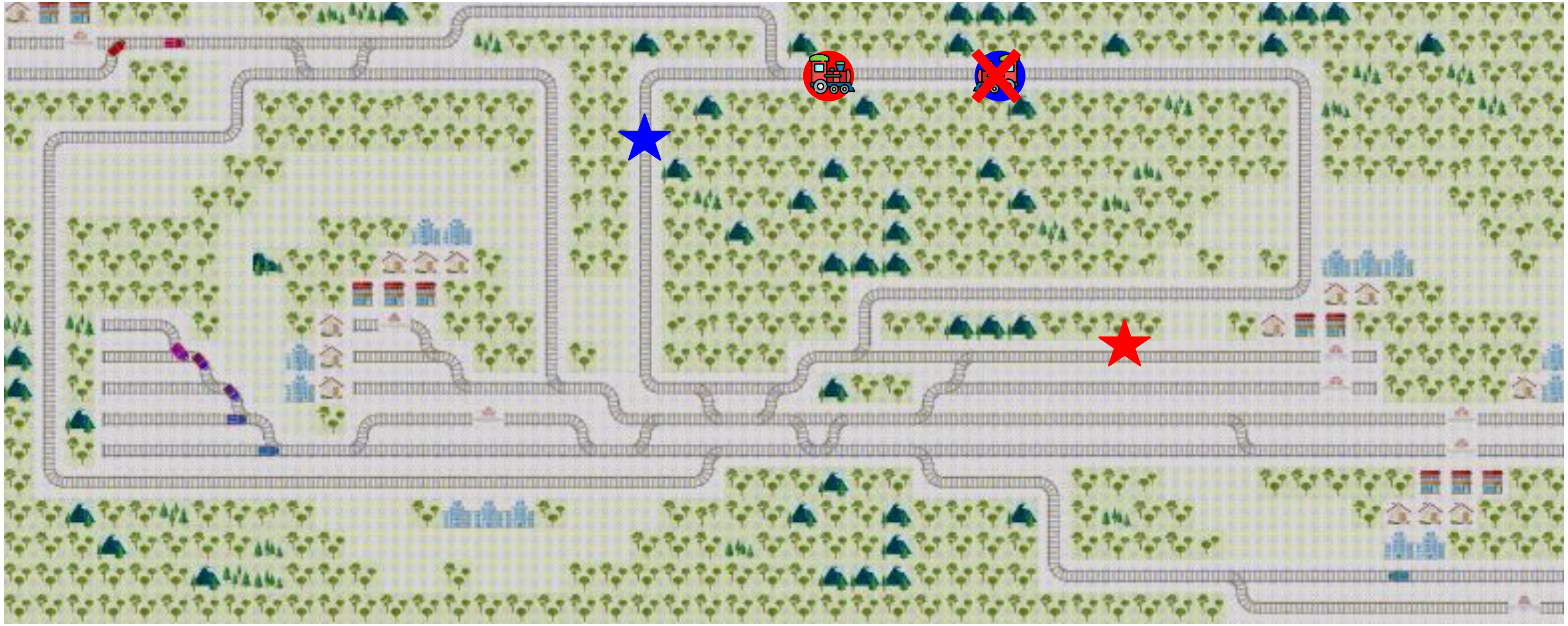

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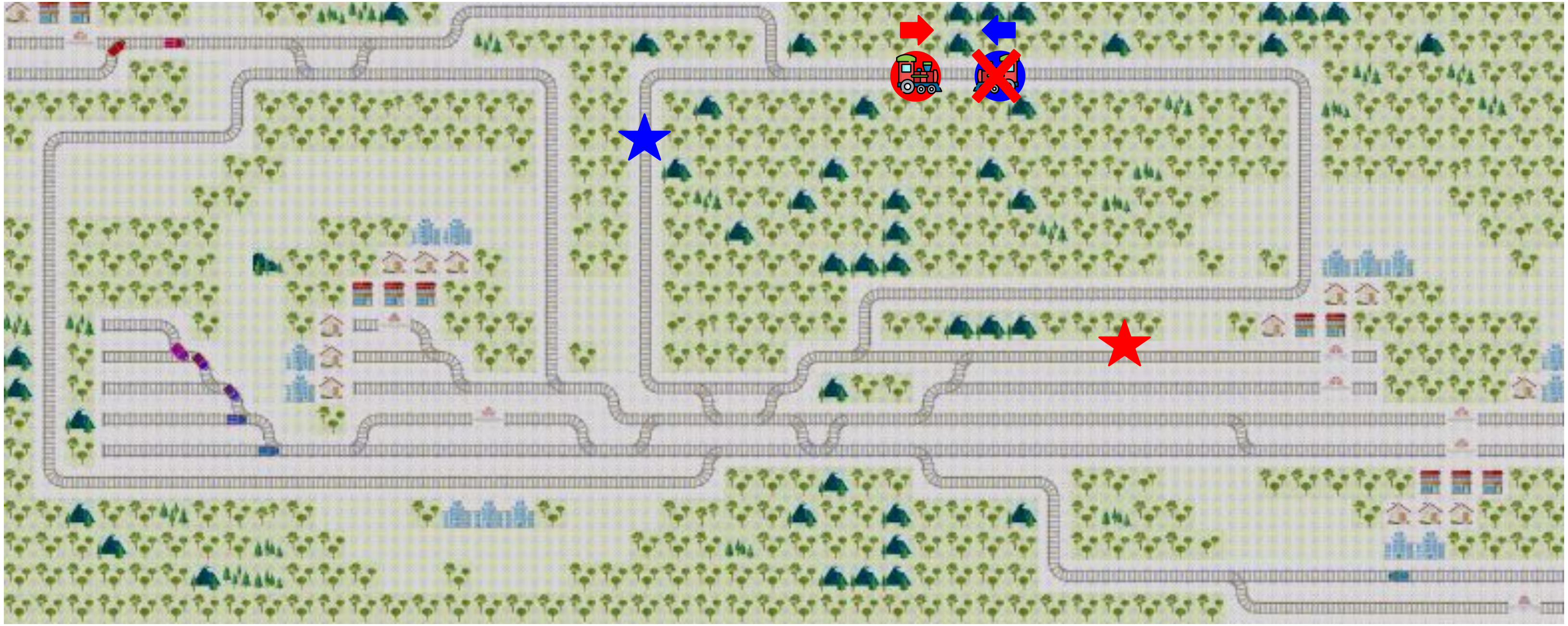

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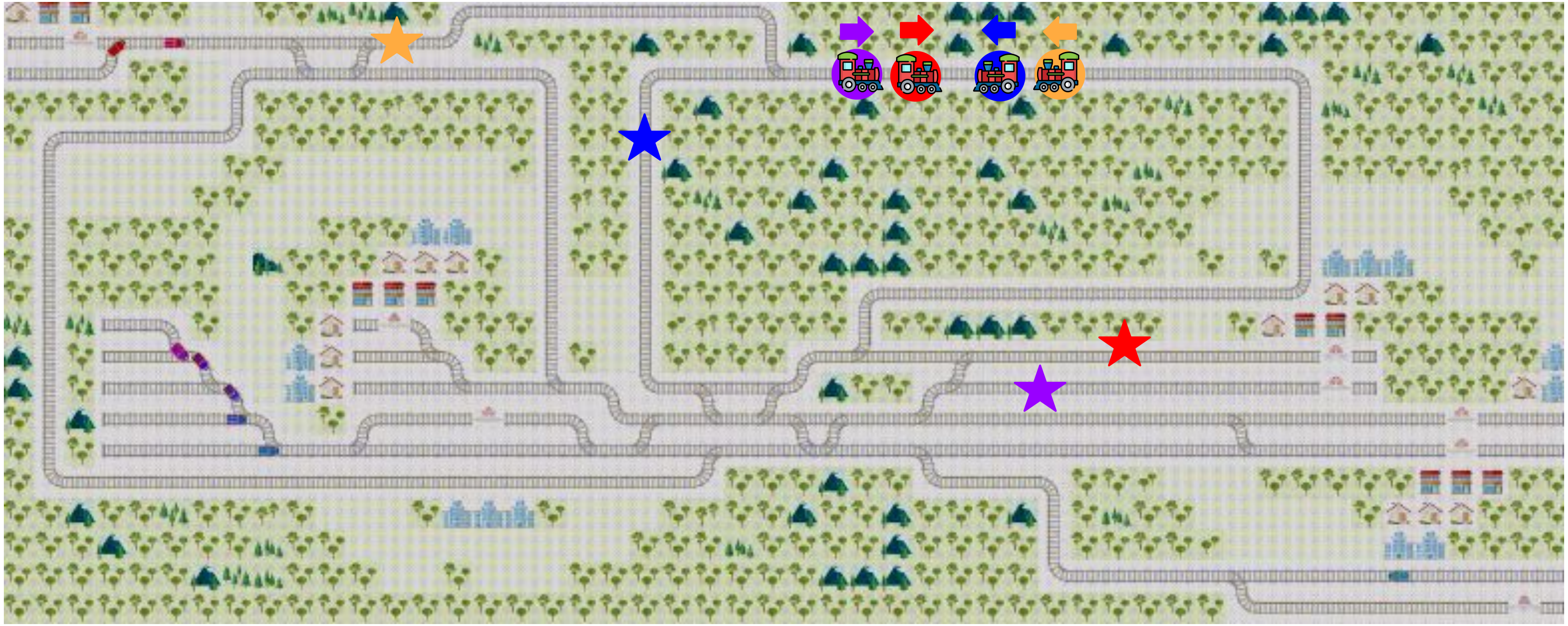

Worst nightmare = Deadlocks

The mission

How to solve this situation ? **Flatland** expert solvers exist but requirements:

- global view of the railroad system
- perfect knowledge of trains status

$\Big\}$ **Global policy**

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Global policy

Reality :

- latency in communications
- train company vs infrastructure company

We don't know where the train is

The mission

How to solve this situation ? **Flatland** expert solvers exist but requirements:

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Global policy

Reality :

- latency in communications
- train company vs infrastructure company

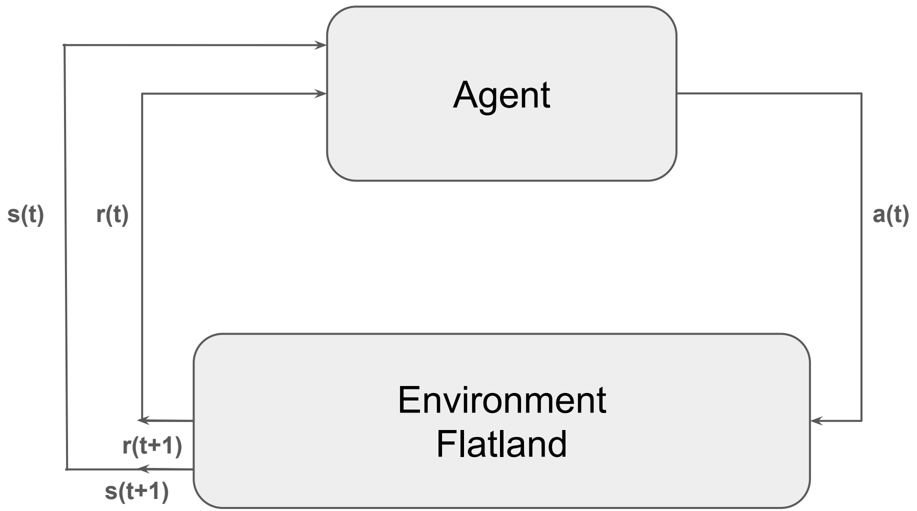
We don't know where the train is

Transfer the choice of action to the train driver → Local policy

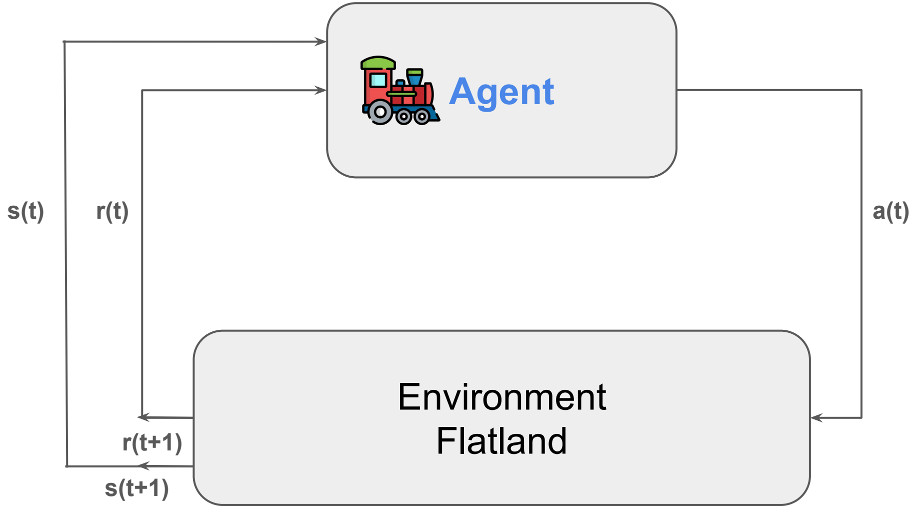
Our contributions

- Transfer global knowledge to local policy with World Model and Imitation learning
- Fine tune local policy with reinforcement learning
- Combining both → new AI SOTA for public Flatland benchmark (+23%)
- Good performances on out-of-distribution instances

Reinforcement Learning

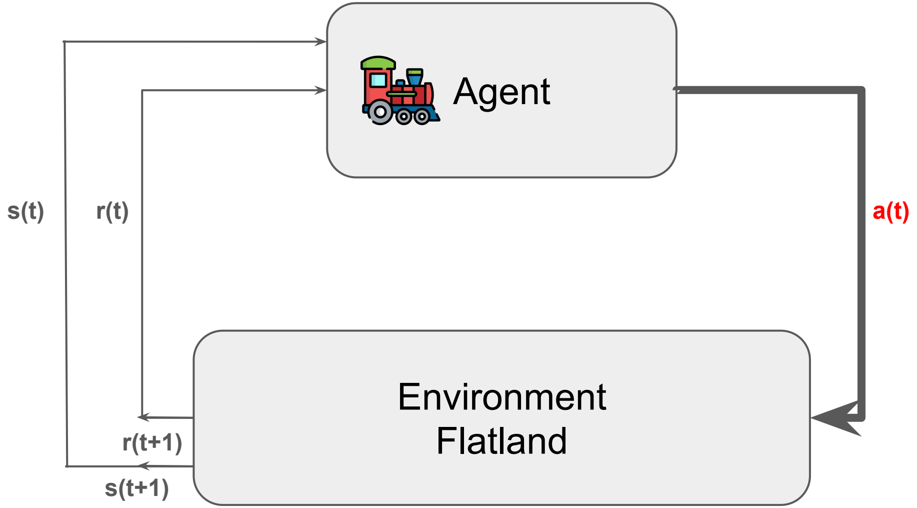


Reinforcement Learning



Agent : train

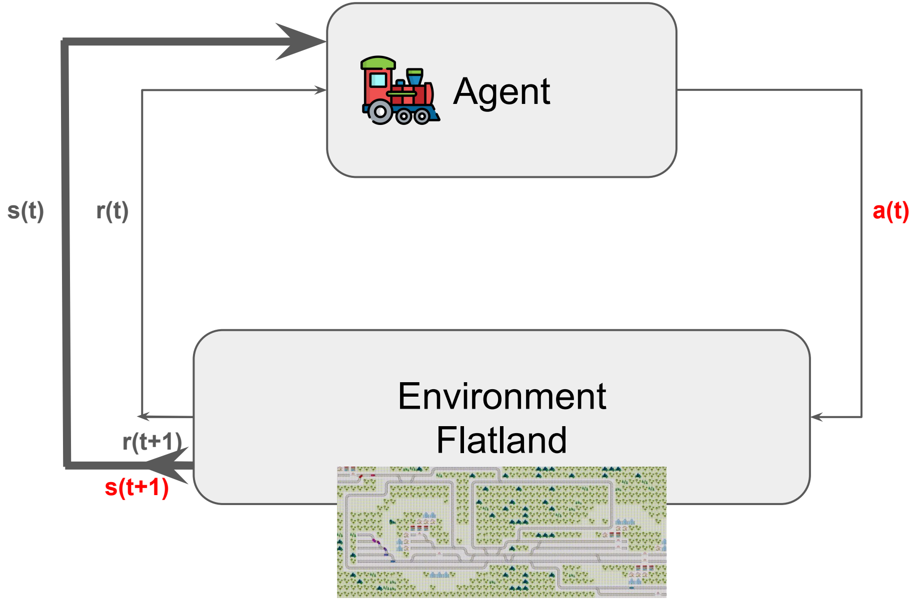
Reinforcement Learning



Agent : train

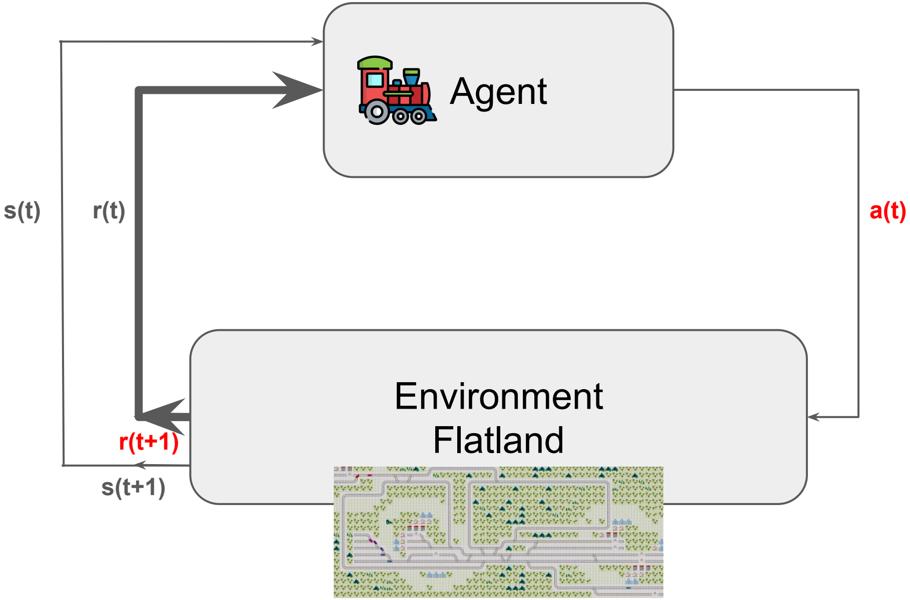
Action : Left, Right, Forward, Stop

Reinforcement Learning



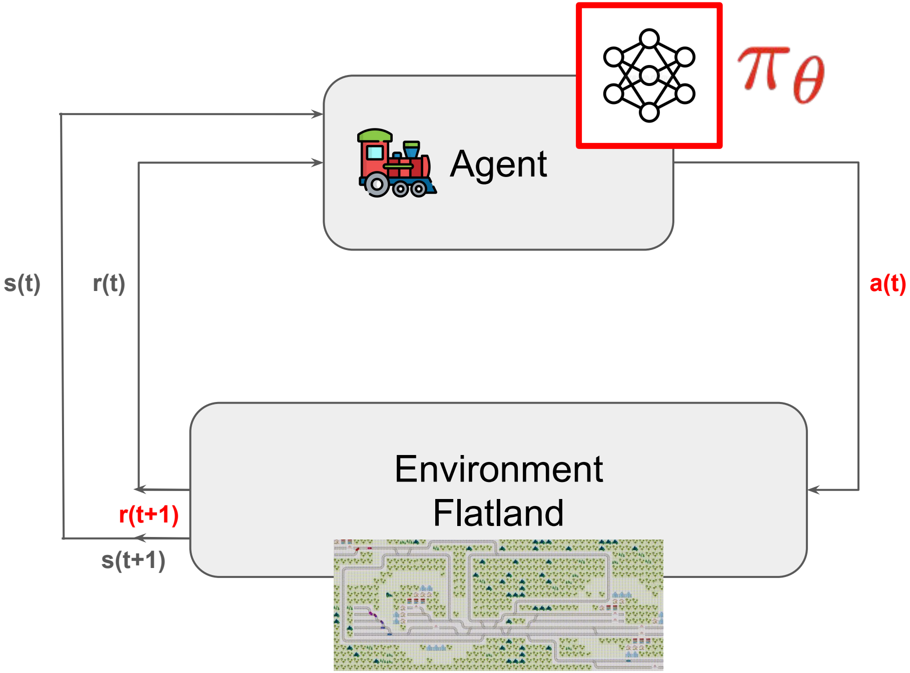
Agent : train
Action : Left, Right, Forward, Stop
State : observation of environment

Reinforcement Learning



Agent : train
Action : Left, Right, Forward, Stop
State : observation of environment
Reward : closer to destination

Reinforcement **Learning**



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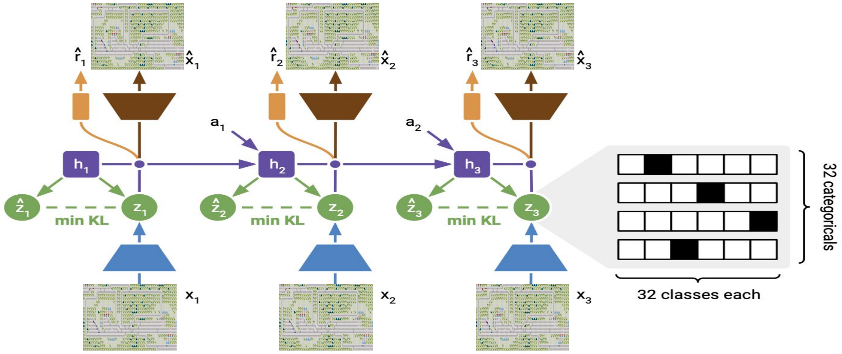
Reward : closer to destination

Model Based Approach - MAMBA

1st training phase : World Model

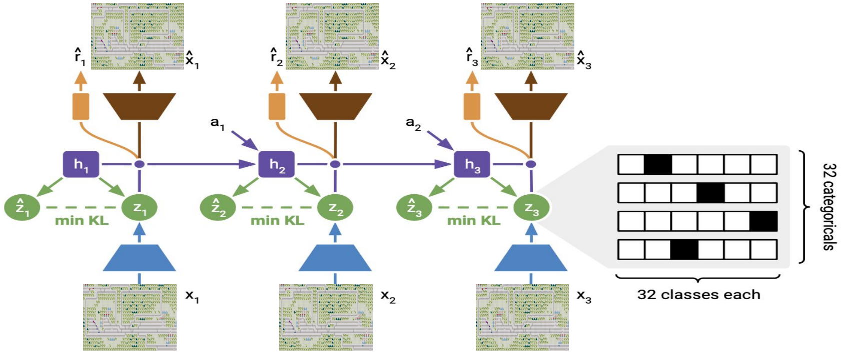


world model = create latent state

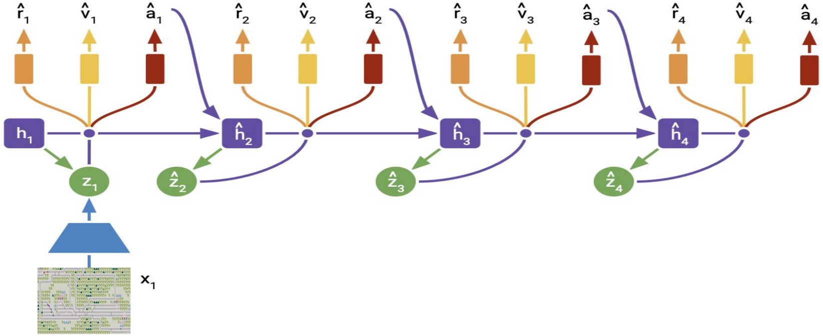
MASTERING ATARI WITH DISCRETE WORLD MODELS, D.Hafner et al. 2021

Model Based Approach - MAMBA

1st training phase : World Model



2nd training phase : PPO Agent



world model = create latent state

agent = best action

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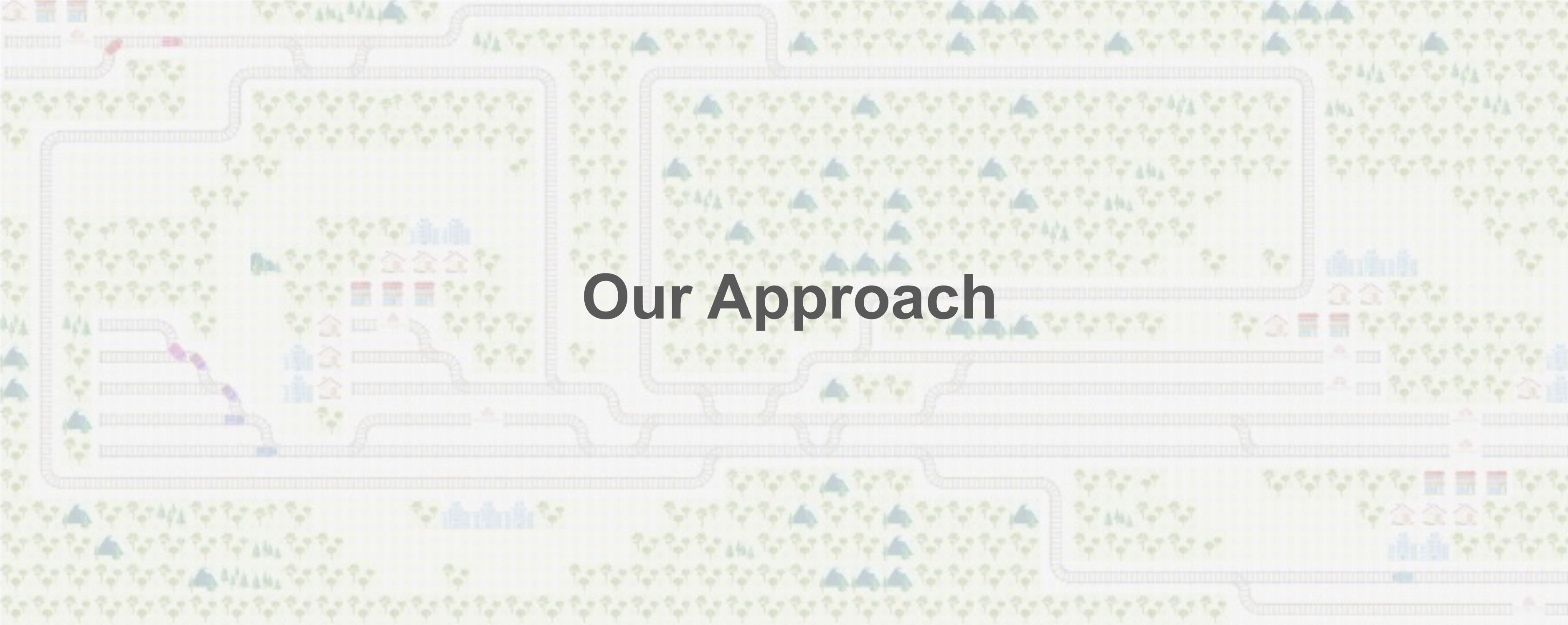
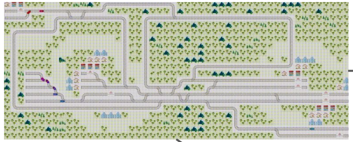


Our Approach

General Architecture

We have access to Expert actions ⇒ Imitation Learning



Expert

Actions a*

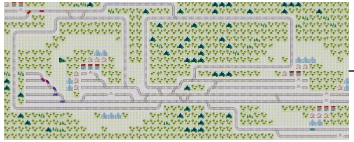
?

World Model

State
$(\hat{z}_t, h_t)$

General Architecture

We have access to Expert actions → Imitation Learning



Expert → Actions a*

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State $(\hat{z}_t, h_t)$ →

Agent

→ Actions

General Architecture

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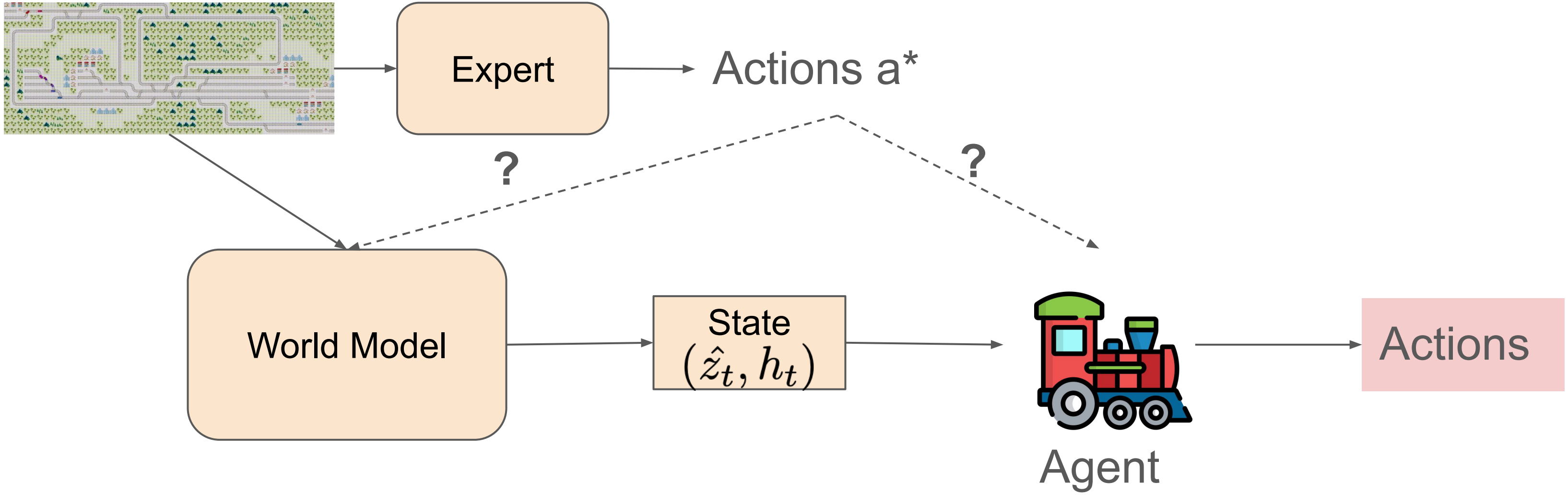
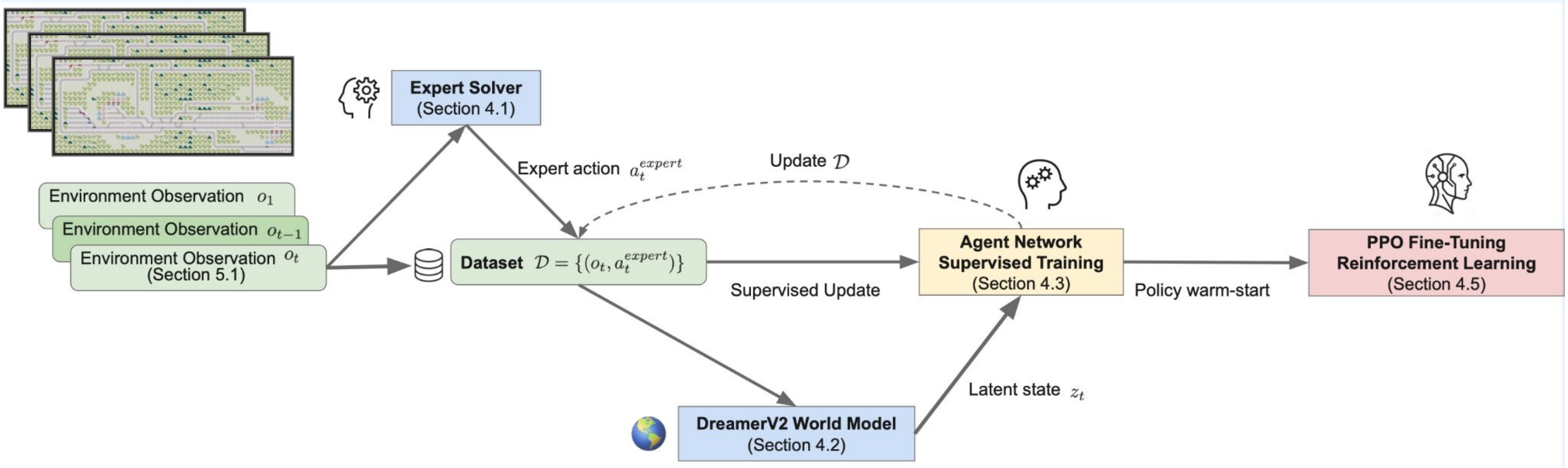


Our Approach : Sequential



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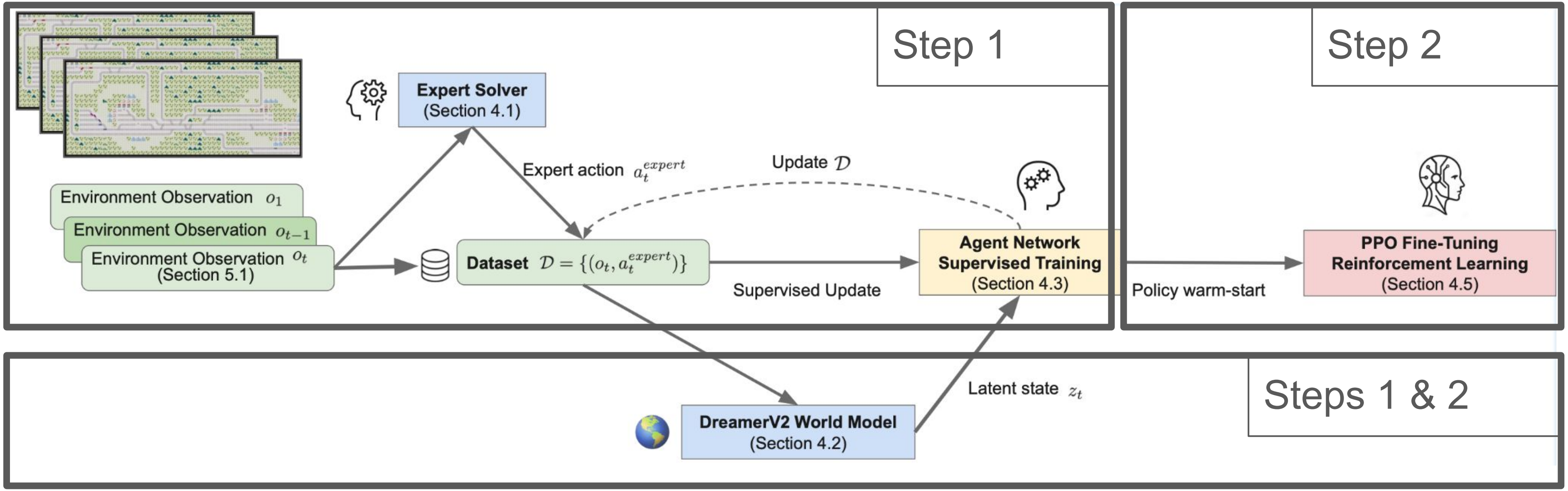

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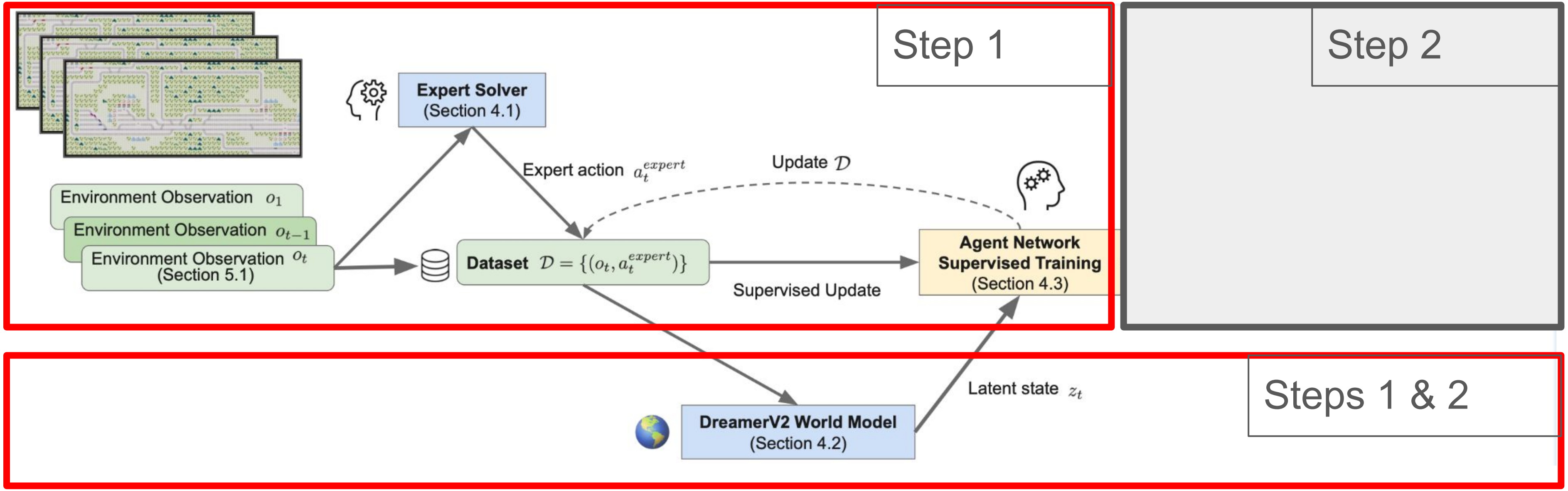
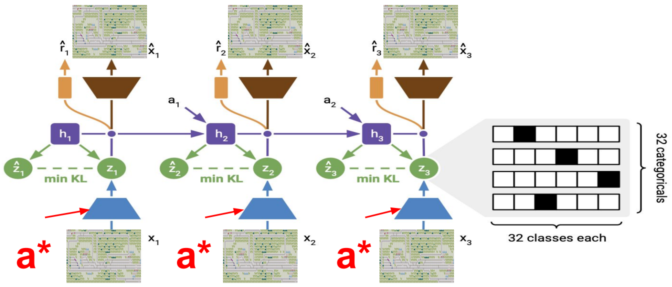


Our Approach : Sequential

Train world model with **a***



Model-Based Imitation Learning for Urban Driving, A.Hu et al. 2022

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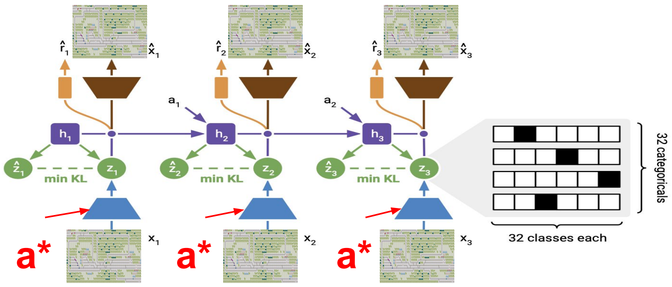

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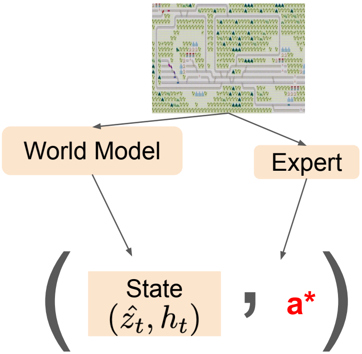
Collect trajectories following current policy



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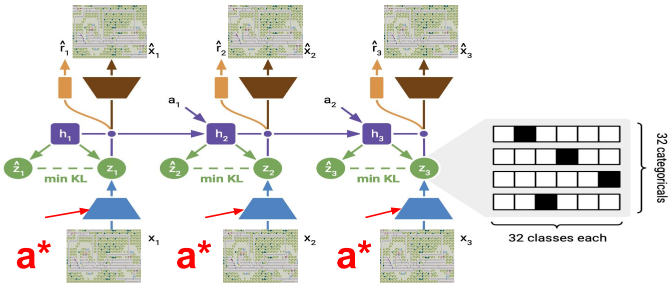

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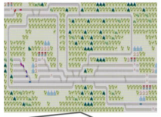


World Model

Expert

(State $(\hat{z}_t, h_t)$, **a***)

Update policy

State $(\hat{z}_t, h_t)$ →  → Actions â

[0.1,0.4,0.2,0.3]

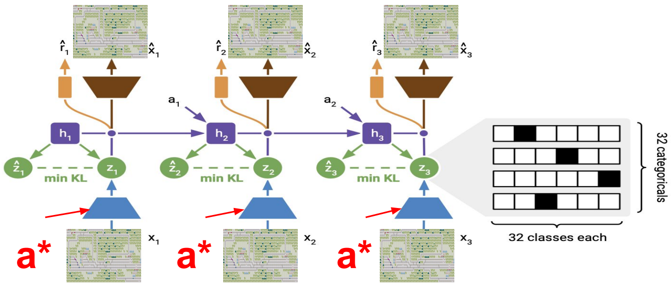
a* → $\mathcal{L}_{IL}$

[0,1,0,0]

$$\mathcal{L}_{IL} = -\sum_{c=0}^{3} a_c^* log(\hat{a}_c)$$ Cross entropy loss

Our Approach : Sequential

Train world model with **a***



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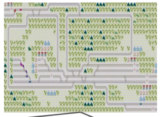

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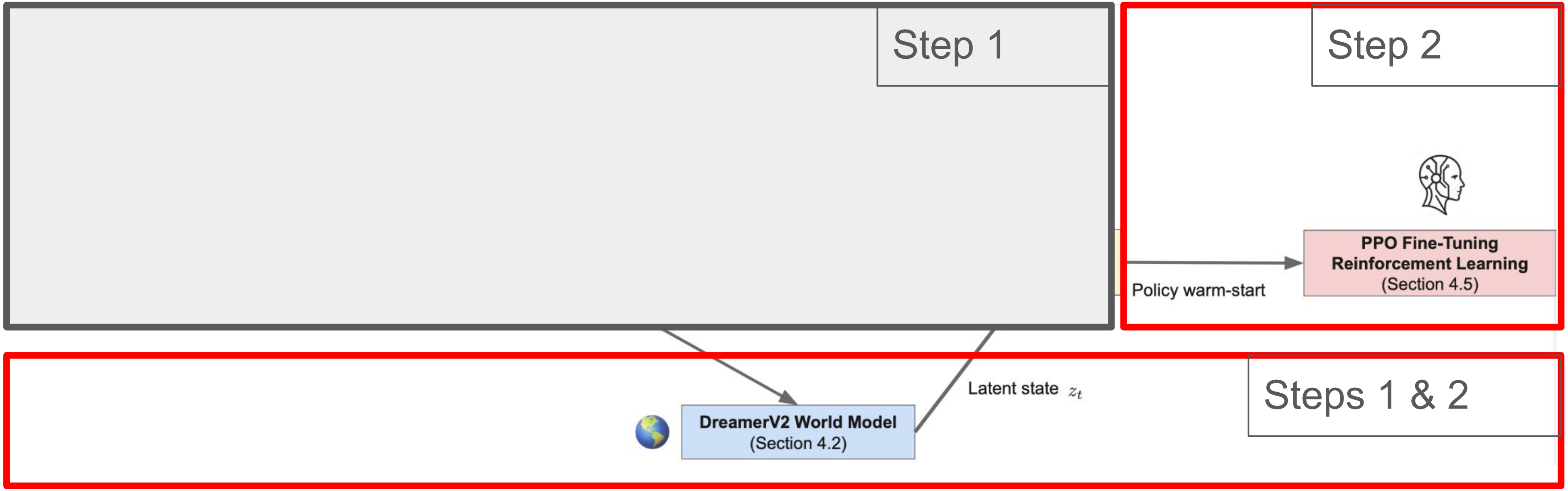
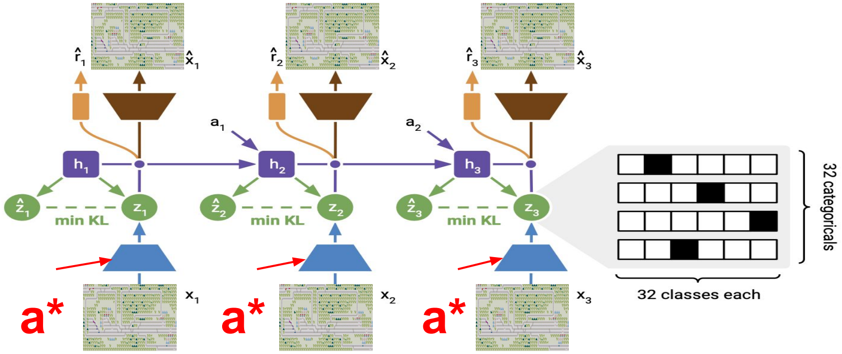
Repeat until convergence

Our Approach : Sequential



Our Approach : Sequential

Continue training world model with **a***



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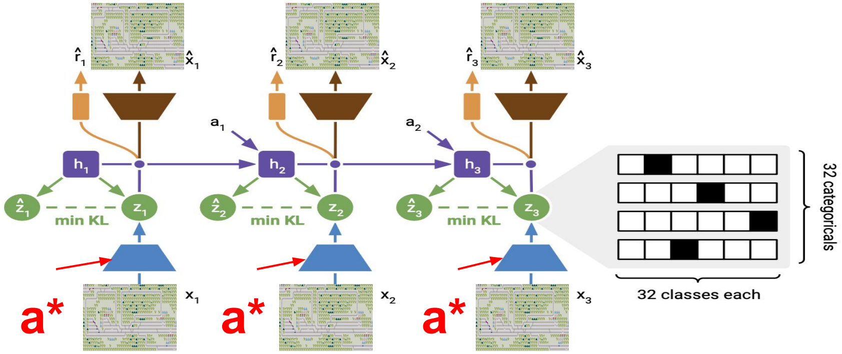

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Initialize policy with step 1 weights

Start RL training
PPO Algorithm

Experimental setup

Policies:

- Random: next action at random among valid actions
- Greedy: Shortest path action
- Expert solver: Prioritized Plan. + LNS + Safe interval path plan. + MCP + SA
- SOTA AI 10: MAMBA trained only on 10-trains instance (220h training)
- SOTA AI++: MAMBA trained on same distribution as test instances(3x220h training)
- Our: Imitation Guided World model trained on 10-trains instances (220h training)

Test on instances : 5 trains 10 trains 15 trains

Main Results

	5 trains		10 trains		15 trains	
Policy	% done	% blocked	% done	% blocked	% done	% blocked
Random	25.0 ± 0.8	63.0 ± 0.8	11.8 ± 0.9	80.7 ± 0.9	8.2 ± 1.1	84.2 ± 1.1
Greedy	18.4 ± 0.8	55.0 ± 0.8	10.7 ± 0.9	79.1 ± 0.9	8.5 ± 1.1	81.3 ± 1.1
Expert solver	**100.0 ± 0**	**0 ± 0**	**98.0 ± 1.1**	**1.8 ± 1.0**	**98.1 ± 1.4**	**1.5 ± 1.1**
SOTA AI 10	48.4 ± 0.6	0.4 ± 0.2	66.1 ± 1.9	**0.2 ± 0.1**	30.9 ± 1.7	**1.6 ± 0.5**
SOTA AI++	97.2 ± 0.6	2.0 ± 0.5	66.1 ± 1.9	**0.2 ± 0.1**	47.6 ± 2.4	4.6 ± 1.7
Our	**98.2 ± 0.3**	**0 ± 0**	**80.8 ± 2.2**	3.2 ± 0.6	**71.3 ± 3.4**	7.7 ± 1.8

Huge increase in performance in difficult instances

% done = f(Breakdown probability)

Policy	Breakdown probability	1/10	1/50	1/100	1/500	1/1000	1/10000	0
SOTA AI++		**94.8**	**99.2**	97.2	98.8	99.2	97.6	**99.2**
Our		50.6	96.6	**98.2**	**99.0**	**100.0**	**100.0**	**99.2**

Policy	Breakdown ratio	1/10	1/50	1/150	1/500	1/1000	1/10000	0
SOTA AI++		30.0	53.3	66.1	56.1	58.6	59.1	56.8
Our		**42.3**	**68.2**	**80.8**	**83.7**	**78.9**	**84.7**	**82.7**

Policy	Breakdown probability	1/10	1/50	1/200	1/500	1/1000	1/10000	0
SOTA AI++		30.9	39.3	47.6	48.8	49.5	48.3	47.3
Our		**31.8**	**52.7**	**71.3**	**72.3**	**71.0**	**77.7**	**70.8**

Better generalization to OOD instances

Ablation Study

	5 agents		10 agents		15 agents	
Policy	% done	Δ	% done	Δ	% done	Δ
Our	**98.2**	-	**80.8**	-	**71.3**	-
With random	75.4	-22.8	76.6	-3.4	58.1	-13.2
With greedy	92.0	-6.2	65.2	-14.8	43.7	-27.6
No world model	0	-98.2	30.0	-50.8	0	-71.3
Pure imitation	2.8	-95.4	0.4	-80.4	0	-71.3

All components are essential

Take-home messages

- Railway networks are **hard to solve**
- **World Models** are promising
- Can we improve them with **expert knowledge** ? ⟶ **YES** !
- Expert knowledge **embedded in state**
- Local agent trained to **mimic global expert** behaviour